NumXventure: A Gamified Learning Application for Improving Proficiency in Numerical Expressions Among Selected Grade 8 Learners in a Public Secondary School in Mayorga, Leyte

PROPOSED INNOVATION, INTERVENTION, AND STRATEGY

This study proposes NumXventure, an online, gamified, team-based learning application designed to improve Grade 8 students' proficiency in numerical expressions, particularly in basic operations, exponential notation, and order of operations. The application integrates interactive problem-solving tasks, instructional review materials, real-time scoring, and collaborative gameplay mechanics within a structured digital environment.

Learners engage in a series of progressively challenging stations where they solve mathematical problems, earn or lose points (gems), and contribute to their team's overall performance. In Station 1 (Basic Operations), students strengthen their fluency and conceptual understanding of integers through repeated practice in performing addition, subtraction, multiplication, and division, with immediate feedback that helps correct misconceptions, especially in handling positive and negative numbers. In Station 2 (Exponential Notation), students build conceptual understanding of exponents as repeated multiplication through guided exercises that involve interpreting, simplifying, and evaluating exponential expressions. In Station 3 (Order of Operations), learners develop procedural accuracy and logical sequencing skills by consistently applying the correct order of operations, reinforced through structured problem-solving tasks that discourage random computation.

To support preparation and conceptual readiness, the system includes a "Learn" feature that provides instructional materials, examples, and guided explanations aligned with the three competencies. However, this feature is only accessible prior to gameplay, serving as a structured review tool before learners enter the adventure phase. During actual gameplay,

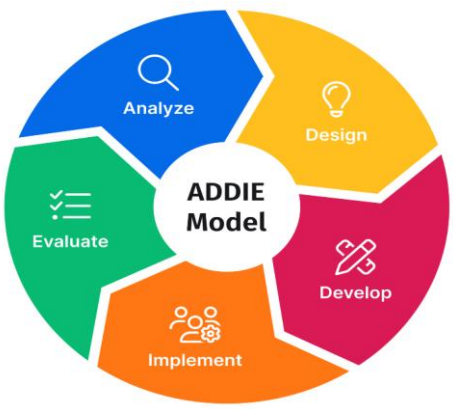
students are expected to rely on their prepared understanding, encouraging independent application of knowledge under timed and gamified conditions. This design ensures that learning support is provided in advance without interrupting the cognitive demands and authenticity of the game-based assessment environment.

The system integrates essential features such as a player dashboard, structured learning modules, leaderboard rankings, and support tools including hints, skip, and rescue options. The hint system provides scaffolded assistance to guide learners step-by-step during problem-solving, while the skip feature allows students to move past an item they find temporarily overwhelming without interrupting the flow of the activity. Additionally, the rescue option fosters peer collaboration by allowing a teammate to transfer gems to assist one member. This feature not only promotes cooperative learning within the game environment but also encourages students to support one another as they work toward a common team goal. Through this design, NumXventure promotes active participation, immediate feedback, strategic thinking, and collaborative learning. Ultimately, the intervention aims to strengthen foundational mathematical skills, address learning gaps, and increase student motivation and engagement through an immersive and meaningful learning experience.

The development and implementation of the NumXventure application will be guided by the ADDIE Model, as shown in Figure 1, to ensure a systematic and structured instructional design process.

Figure 1.

The ADDIE Instructional Design Model



Note. From “How to Use the ADDIE Instructional Design Model”, by D.R., Moiso, 2024, SessionLab. (https://www.sessionlab.com/blog/addie-model-instructional-design/). Copyright 2024 by Moiso.

Analyze

The Analysis phase involves identifying learners’ needs, defining existing learning problems, and determining the instructional goals that the proposed intervention must address. This phase serves as the foundation for the design and development of an effective and responsive learning solution.

The target participants of this study are Grade 8 learners. Based on the results of the Rapid Mathematics Assessment (RMA) conducted at the end of School Year 2025–2026, a significant number of learners have not achieved the expected level of proficiency in numerical expressions. Out of the total population of 248 Grade 8 learners, 2 were classified as not proficient, 24 as low proficient, and 66 as nearly proficient. These findings indicate that a considerable proportion of learners are performing below the expected standard, despite the fact that proficiency in numerical expressions is a competency that should have been developed and mastered by the end of Grade 7.

Further analysis of learners’ responses reveals specific areas of difficulty. Although most learners demonstrate basic knowledge of addition, subtraction, multiplication, and division, they exhibit misconceptions when dealing with negative numbers, particularly in applying operational rules involving positive and negative signs. This suggests gaps in conceptual understanding rather than a complete lack of procedural knowledge.

In addition, learners show challenges in understanding and applying exponential notation. Difficulties include writing expressions in exponential form, interpreting exponents correctly, and performing operations involving powers. These challenges suggest limited conceptual understanding of exponents and their relationship to repeated multiplication.

Moreover, learners experience difficulty in applying the correct order of operations when solving numerical expressions. Many learners fail to apply the prescribed order of operations correctly, resulting in incorrect solutions, particularly when solving numerical expressions involving multiple operations. This indicates a need for structured guidance and repeated practice in organizing and simplifying expressions systematically.

Additionally, it was observed that most learners have access to mobile phones and frequently use them during their free time. Many engage in mind games applications, such as Word Riddle and Brain out, indicating familiarity with and interest in. This suggests that a mobile, game-based learning approach is both accessible and appropriate, as it aligns with learners' existing habits and can enhance engagement with the NumXventure application.

Given these identified learning gaps in basic operations with integers, exponential notation, and order of operations, there is a clear need for an intervention that provides structured, scaffolded, and engaging learning experiences. Thus, the NumXventure application is designed to directly address these competencies by offering interactive tasks, immediate feedback, and opportunities for practice and collaboration, ultimately aiming to improve learners' proficiency in numerical expressions.

Design

This phase focuses on systematically planning the structure, content, features, and user interaction of the learning application. It defines how instructional objectives will be translated into engaging and meaningful learning experiences, including the organization of content, interface layout, game mechanics, and assessment strategies.

For the NumXventure application, the design will be carefully structured to align with the target competencies in numerical expressions while promoting engagement through gamification and collaboration. The application will be designed as a team-based adventure game, where learners will be grouped into teams consisting of five to six members. This

structure is intended to encourage shared responsibility, peer support, and collaborative problem-solving, consistent with Constructivist Learning Theory.

The application will accommodate two types of users: students and teachers (admin). Students will be required to create accounts, log in to the system, and choose their characters to access the game features. Meanwhile, teachers will be provided with an Admin Dashboard, which will serve as the control and management center of the application. Through the Admin Dashboard, teachers will be able to input, edit, and organize questions per station; monitor individual and team performance; set and adjust time limits; and control the flow of the game, including starting and pausing the activity.

The student interface will include a Player Dashboard that will provide access to four primary features:

a. **Home**, which will serve as the default landing page of the student interface, displaying the learner's username and providing a logout button in the upper right corner to support user identification, secure account access, and efficient navigation within the application.
b. **Start Adventure**, which will serve as the main gameplay environment where learners solve mathematical problems;
c. **Leaderboard**, which will display team rankings based on accumulated scores to foster motivation and healthy competition; and
d. **Learn**, which will function as a review hub containing instructional materials on basic operations, exponential notation, and order of operations.

The gameplay will be organized into three progressive stations; each aligned with specific competencies:

- Station 1: Basic Operations (20 items)
- Station 2: Exponential Notation (10 items)

- Station 3: Order of Operations (10 items)

A central component of the design will be the Gems Mechanics, which will serve as the scoring, feedback, and motivational system of the application. Each player will begin with 50 gems, which will increase or decrease based on their performance and decisions. At the end of the allotted adventure time, the total gems of all members within each team will be combined, and the team with the highest accumulated gems will be declared the winner. This mechanism is intended to reinforce collective responsibility, encourage strategic collaboration, and promote active participation among all team members.

The scoring system will be structured as follows:

A. Item Scoring

- Correct Answer: +10 gems
- Incorrect Answer: −5 gems
- Use of Hint: −5 gems
- Skip Item: −15 gems
- Perfect Completion of a Station: +10 bonus gems

B. Group Score Calculation

- Team Final Score = sum of the gems of all members
- Eliminated Member (reaching 0 gems): −50 gems
- Member who will not complete all stations within the time limit: −25 gems

C. Rescue System

- A teammate will be allowed to transfer 10 gems to assist one member only

These mechanics will be grounded in Operant Conditioning, where positive behaviors will be reinforced through rewards, while undesirable responses will be discouraged through appropriate consequences. The design is intended to balance motivation, accountability, and strategic decision-making.

The +10 reward for correct answers is expected to reinforce mastery, while the −5 penalty for incorrect answers will encourage learners to think carefully before responding. The −5 deduction for hints will support scaffolding without discouraging help-seeking, whereas the −15 penalty for skipping items will discourage avoidance and promote perseverance. The bonus for perfect completion is intended to motivate consistency and accuracy.

At the group level, penalties such as −50 gems for eliminated members and −25 for incomplete participation are expected to emphasize accountability and time management. Meanwhile, the rescue system will introduce a collaborative element, allowing learners to support teammates who are at risk, thereby promoting teamwork, empathy, and shared responsibility.

Overall, the design of NumXventure is intended to integrate instructional content, game mechanics, user management features (including the Admin Dashboard), and collaborative structures to promote engagement, critical thinking, and sustained participation among learners.

Develop

The Development phase involves the creation and assembly of the learning application based on the established design specifications. This phase includes building the interface, integrating content, developing system functionalities, and conducting testing procedures to ensure usability, effectiveness, and alignment with learning objectives.

For NumXventure, the development process will involve constructing the overall application framework, including the Player and Admin Dashboards, gameplay environment, leaderboard, and learning modules. The interface will be designed to be simple, intuitive, and user-friendly in alignment with Cognitive Load Theory to minimize unnecessary cognitive load during learning.

The application will be developed by programming its core features and mechanics using appropriate software development tools, ensuring that all components—such as navigation, scoring systems, timers, and interactive elements—function accurately and efficiently. Validated learning materials and assessment items will be integrated and organized according to the three stations, allowing learners to progress from basic to more complex tasks. Interactive features such as Hint, Skip, and Rescue will be embedded and tested to ensure proper functionality and responsiveness.

The Gems Mechanics will be programmed to automatically compute scores, track remaining gems, and update team rankings in real time based on learners' actions during gameplay. In addition, built-in data tracking features will be developed to record learners' performance, including number of correct answers and accumulated gems, which will serve as preliminary indicators of learning progress and engagement.

To ensure the quality and effectiveness of the application, the system will undergo rigorous testing and refinement, including debugging, pilot testing, and usability checks. Feedback gathered during pilot testing will be used to improve system functionality, clarity of instructions, and overall user experience. These processes will also serve as an initial evaluation of the intervention by identifying technical issues, usability concerns, and potential learning barriers before full implementation.

Implement

The Implementation phase involves delivering and facilitating the use of the developed application among the target learners. It focuses on how the application is introduced, accessed, and utilized in an actual learning setting.

In this study, the implementation of NumXventure will be conducted with the identified Grade 8 learners who demonstrated below the expected proficiency level in numerical expressions. Prior to the intervention, students will be oriented on how to use the application,

including account registration, navigation of the dashboard, and understanding of the game mechanics, rules, and objectives.

Learners will then be grouped into teams consisting of five to six members and provided with a team name to access the game. During each session, students will log into the application and participate in the Adventure Phase, where they will answer problems across the three stations while managing their gems and contributing to their team's score. Within the application, students will solve items sequentially per station, decide strategically whether to answer, use hints, or skip items, monitor their remaining gems and time, and collaborate with teammates through the rescue system when needed.

The teacher will facilitate the process by using the Admin Dashboard to monitor progress, control time, and ensure that the activity runs smoothly. Real-time data from the application will allow the teacher to identify learners who may need additional support during gameplay.

Through this structured implementation, learners actively engage with the application, experience immediate feedback, and participate in collaborative and competitive learning, consistent with Experiential Learning Theory, where knowledge is developed through active involvement and reflection.

Evaluate

This phase focuses on determining the effectiveness, usability, and overall impact of the learning application. It involves analyzing both performance data and user feedback to assess whether the application meets its intended objectives and to identify areas for improvement.

For NumXventure, evaluation will utilize multiple sources of data. These include in-app performance indicators such as accumulated gems, correct responses, and team rankings.

These quantitative data provide insights into learners' progress, engagement levels, and mastery of the targeted mathematical skills.

In addition, learners' perceptions of the NumXventure application will be assessed using a structured survey questionnaire administered after the implementation phase. The instrument will gather data on learners' motivation, engagement and enjoyment, perceived usefulness in improving mathematical skills, and overall learning experience.

The combination of performance data and learners' responses ensures a comprehensive evaluation of NumXventure, covering both learning outcomes and user experience. This approach allows for a more holistic understanding of the application's effectiveness in terms of usability, engagement, and acceptability. The findings from this evaluation will serve as a basis for necessary revisions and improvements, ultimately enhancing the application's capacity to support learners' proficiency in numerical expressions.

The NumXventure intervention is grounded on Constructivist Learning Theory, Experiential Learning Theory, Operant Conditioning, and Cognitive Load Theory, which collectively explain how students develop proficiency in numerical expressions within a structured gamified environment. Constructivist Learning Theory emphasizes that learners construct knowledge through active engagement with problems and collaboration with peers, which is reflected in the team-based adventure gameplay. Experiential Learning Theory supports the cycle of concrete experience, reflection, and application as students solve tasks, receive immediate feedback, and adjust their strategies across stations. Operant Conditioning Theory explains how the gem-based reward system reinforces correct responses and productive learning behaviors while discouraging errors, guessing, and task avoidance through structured consequences. Cognitive Load Theory supports the intentional separation of the "Learn" feature from gameplay, ensuring that instructional materials are used for pre-learning preparation rather than during active problem-solving, thereby reducing cognitive overload and

promoting retrieval-based application during the game. Collectively, these theories provide a strong foundation for NumXventure as an intervention that enhances conceptual understanding, procedural fluency, learner independence, and sustained engagement in mastering numerical expressions.